More frequent and intense convective downdrafts with warming in km-scale simulations

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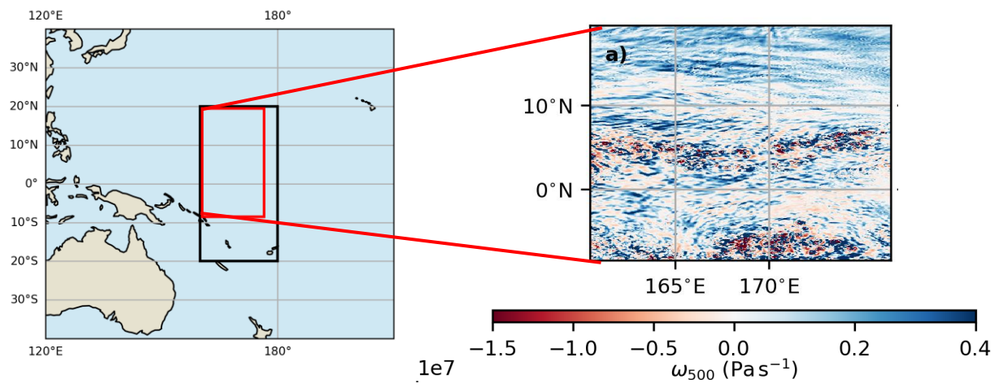
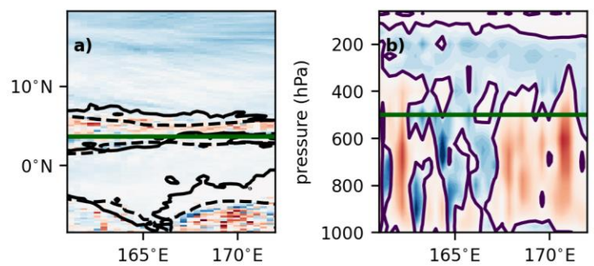
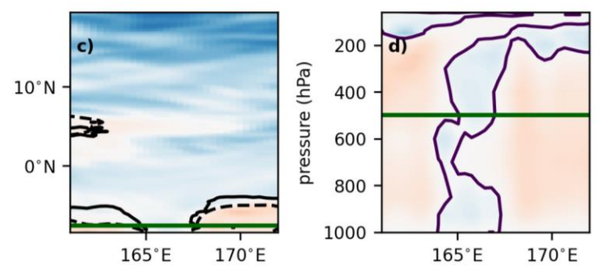
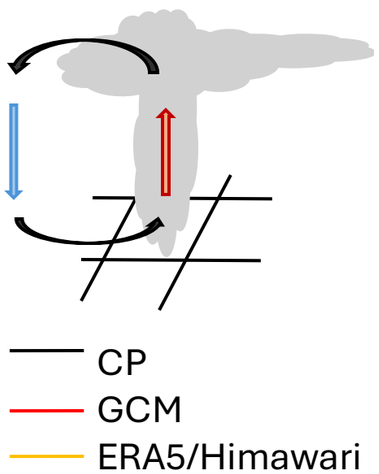
1. How is cloud-circulation coupling different when using a convection-permitting simulation?

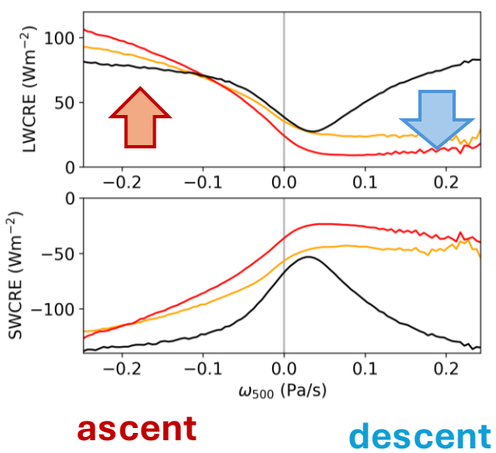


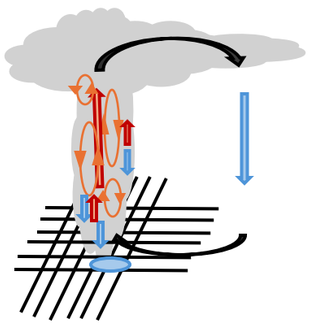




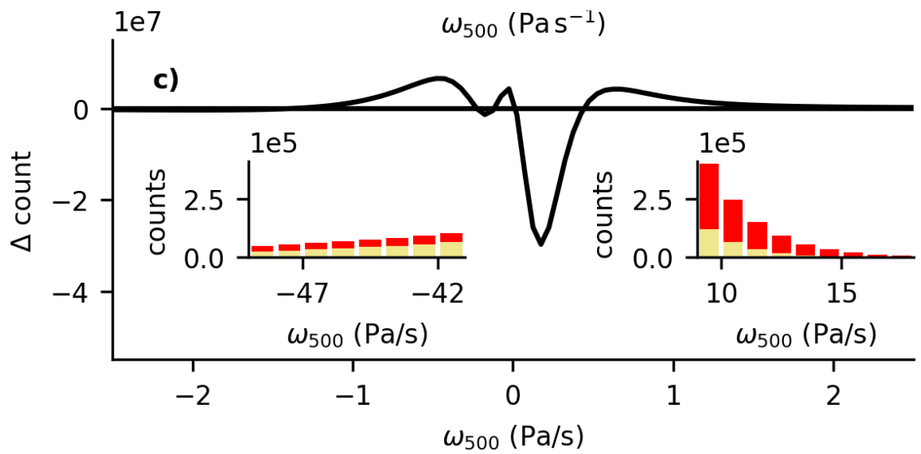
2. Complex updraughts and downdraughts in convective core alter fundamental cloud-circulation relationship: presence of downdraughts reduces the cooling effect of clouds







3. With warming, we find an increase in frequency of downdraughts.



Increase in convection in moister regions → **Increase in ascent rate in moister regions** → **Increase in condensate in convective core** → **Increased condensate loading leads to more frequent downdraughts**

Mackie et al., submitted